X-CAVATE Protocol

G-Code Generation from Vascular Network Geometry

Stage II of the Vascular Network 3D Bioprinting Workflow

This protocol describes how to use X-CAVATE to convert SimVascular b-spline output into collision-free 3D printer G-code. X-CAVATE can be run in three modes: a **web GUI** (recommended for new users), a **command-line interface** (CLI), or via **Docker**. The steps below cover all three.

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1 Part A: Installation (5 minutes)

1.1 Option 1: GUI or CLI (pip install)

1. Open a terminal (Mac: Terminal.app; Windows: Anaconda Prompt or PowerShell).
2. Confirm Python 3.9+ is installed:

```
python3 --version
```

3. Navigate to the folder where you want to store X-CAVATE and clone the repository:

```
git clone https://github.com/skylarscottlab/X-CAVATE.git
cd X-CAVATE
```

   Alternatively, download and unzip the repository from the GitHub page (green "Code" button → "Download ZIP").

4. Install X-CAVATE with GUI support (includes all required libraries: numpy, scipy, pandas, plotly, streamlit):

```
pip install ".[gui]"
```

   If you only need the CLI (no web interface):

```
pip install .
```

5. Verify the installation:

```
xcavate --help
```

   You should see the full list of command-line parameters.

1.2 Option 2: Docker (no Python required)

1. Install Docker Desktop from https://www.docker.com/products/docker-desktop.
2. Open a terminal and navigate to the X-CAVATE folder:

```
cd X-CAVATE
```

3. Build the Docker image:

```
docker build -t xcavate .
```

4. Run the container:

```
docker run -p 8501:8501 xcavate
```

5. Open your browser to **http://localhost:8501**. The X-CAVATE web interface should appear.

2 Part B: Prepare Input Files (5 minutes)

X-CAVATE requires two input files from your SimVascular workflow.

2.1 B.1 Network File (b-splines)

This file contains the vessel centerline coordinates exported from SimVascular. Each vessel has a header line followed by coordinate data rows.

```
Vessel: 0, Number of Points: 100
33.4878, 0.0000, 53.5303, 0.0200
33.4115, -0.5285, 52.7311, 0.0200
...
Vessel: 1, Number of Points: 100
15.0600, -37.3300, 11.6600, 0.0190
...
```

- **Columns 1–3** (required): x, y, z coordinates in centimeters.
- **Column 4** (optional): vessel radius, needed if using speed calculation.
- **Column 5** (optional): arterial/venous flag (0 = venous, nonzero = arterial), needed for multimaterial printing.

CAUTION: Coordinates must be in centimeters. X-CAVATE internally converts to millimeters.

CAUTION: Limit each vessel to no more than 100 coordinates to avoid long runtimes.

2.2 B.2 Inlet/Outlet File

This file identifies which network locations are inlets (where printing begins) and outlets.

```
inlet
21.97, -15.43, 50.00
11.81, 0.00, 21.34
outlet
36.05, -27.19, 50.00
50.00, 0.00, 26.54
```

Each section begins with the keyword `inlet` or `outlet` on its own line, followed by one or more (x, y, z) coordinate triplets. You can have any number of inlets and outlets.

Note: Inlet/outlet coordinates are matched to the nearest network point automatically. They do not need to exactly match a coordinate in the network file.

2.3 B.3 File Placement

- **GUI or Docker mode:** Files are uploaded through the web interface. No file placement needed.
- **CLI mode:** Transfer both files into the `inputs/` folder inside the X-CAVATE directory.

3 Part C: Custom G-Code Templates (5 minutes, optional)

Custom G-code templates adapt X-CAVATE's output to your specific printer hardware and software. Each template file contains a G-code snippet that is automatically inserted at the appropriate location in the final G-code output.

The `inputs/custom/` folder contains 12 template files with placeholder text. Replace the placeholder text with your own G-code for each section relevant to your printer.

3.1 C.1 Header

File	Instructions
`header_code.txt`	Replace the placeholder text with your custom G-code for the header section of the final G-code output. Include any code required for the functioning of your printer, such as establishing connections between external pressure boxes and syringes.

3.2 C.2 Single Material Extrusion

Skip this section if printing multimaterial; proceed to C.3.

File	Instructions
`start_extrusion_code.txt`	Replace with your G-code for starting extrusion.
`stop_extrusion_code.txt`	Replace with your G-code for stopping extrusion.

3.3 C.3 Multimaterial Extrusion

Skip this section if printing single material; proceed to C.4.

File	Instructions
`start_extrusion_code_printhead1.txt`	Start extrusion for Printhead 1 (arterial).
`start_extrusion_code_printhead2.txt`	Start extrusion for Printhead 2 (venous).
`stop_extrusion_code_printhead1.txt`	Stop extrusion for Printhead 1 (arterial).
`stop_extrusion_code_printhead2.txt`	Stop extrusion for Printhead 2 (venous).
`active_pressure_printhead1.txt`	Set ink extrusion pressure when Printhead 1 is the **active** printhead.
`active_pressure_printhead2.txt`	Set ink extrusion pressure when Printhead 2 is the **active** printhead.
`rest_pressure_printhead1.txt`	Set ink extrusion pressure when Printhead 1 is the **inactive** printhead.

`rest_pressure_printhead2.txt`	Set ink extrusion pressure when Printhead 2 is the **inactive** printhead.

3.4 C.4 Dwell (Optional)

Incorporating time delays ("dwells") at the first and last node of each print pass deposits additional ink at each junction, improving the connection via a process similar to "spot welding."

File	Instructions
`dwell_code.txt`	Replace with your G-code for creating a time delay at the start and end of each print pass (e.g., `G4 P80` for an 80 ms dwell).

3.5 C.5 How to Provide Custom G-Code in Each Mode

Mode	Instructions
GUI	Enable the "Custom G-code" toggle in the sidebar's Advanced section. A tabbed editor (Header, Single Material, Multimaterial, Dwell) appears in the main area. Paste your G-code into the text areas.
CLI	Edit the `.txt` files in `inputs/custom/` directly with a text editor (e.g., `nano`, TextEdit, Sublime Text), then run with `-custom 1`.
Docker	*Option A:* Use the GUI editor in the browser (no file mounting needed). *Option B:* Mount your local `inputs/custom/` folder: `docker run -p 8501:8501 -v $(pwd)/inputs/custom:/app/inputs/custom xcavate`, then enable "Custom G-code" in the GUI.

4 Part D: Execute X-CAVATE (5 minutes)

4.1 Mode 1: Web GUI

1. Start the server:

   ```
   streamlit run xcavate/gui/app.py
   ```

   A browser window opens at **http://localhost:8501**.

2. In the **sidebar**, upload your Network file and Inlet/Outlet file.
3. Set the **Required Parameters**:
   - **Nozzle diameter** (mm): Outer diameter of the print nozzle.
   - **Container height** (mm): Height of the print container.
   - **Decimal places**: Number of decimal places for output rounding.
   - **Amount up** (mm): Distance to raise the nozzle above the container between passes.
4. Select a **Printer type**: Pressure, Positive Ink, or Aerotech.

5. (Optional) Toggle **Multimaterial** if printing with two inks.
6. (Optional) Expand the optional parameter sections to adjust tolerancing, print settings, geometry, downsampling, multimaterial settings, positive ink displacement settings, and advanced options.
7. If using custom G-code, enable the toggle in the Advanced section and fill in the template editor tabs (see Part C).
8. Click **Run X-CAVATE**.
9. The progress bar tracks the pipeline through 7 steps:
   (a) Reading network and inlet/outlet files
   (b) Interpolating network to nozzle resolution
   (c) Building adjacency graph and detecting branchpoints
   (d) Generating collision-free print passes
   (e) Post-processing: subdivision, gap closure, overlap
   (f) Processing multimaterial passes
   (g) Writing output files and G-code
10. When complete, the main area displays an interactive 3D plot and download buttons.
11. Click the download buttons to save G-code, coordinate files, and the changelog.

4.2 Mode 2: Command-Line Interface

1. Open a terminal and navigate to the X-CAVATE directory:

```
cd X-CAVATE
```

2. (If using custom G-code) Edit the template files in `inputs/custom/` (see Part C), then include `-custom 1` in the command.
3. Run X-CAVATE with your parameters. Example for a single-material pressure print:

```
xcavate \
  --network_file inputs/VesselNetwork.txt \
  --inletoutlet_file inputs/InletsOutlets.txt \
  --nozzle_diameter 0.65 \
  --container_height 50.0 \
  --num_decimals 5 \
  --amount_up 10.0 \
  --printer_type 0 \
  --multimaterial 0 \
  --tolerance_flag 0 \
  --speed_calc 1 \
  --plots 1 \
  --downsample 0 \
  --custom 1 \
```

```
  --flow 0.127
```

Or equivalently: `python -m xcavate [same arguments]`

4. The terminal outputs progress through each pipeline step:
   (a) Network importation
   (b) Interpolation of network coordinates
   (c) Graph generation and branchpoint detection
   (d) DFS/Sweep Line algorithm to generate print passes
   (e) Subdivision and gap closure
   (f) (If multimaterial) Multimaterial pass generation
   (g) (If speed_calc) Print speed calculation
   (h) Output file generation
5. Output files are written to `outputs/` (or `-output_dir`).

Required CLI Parameters

Parameter	Description	Example
`-network_file`	Path to network coordinates file	`inputs/net.txt`
`-inletoutlet_file`	Path to inlet/outlet file	`inputs/io.txt`
`-nozzle_diameter`	Nozzle outer diameter (mm)	`0.65`
`-container_height`	Container height (mm)	`50`
`-num_decimals`	Decimal places for rounding	`5`
`-amount_up`	Z-raise between passes (mm)	`10`
`-printer_type`	0=Pressure, 1=PosInk, 2=Aerotech	`0`
`-multimaterial`	0=single, 1=multi	`0`
`-tolerance_flag`	0=off, 1=on	`0`
`-speed_calc`	0=off, 1=on	`1`
`-plots`	0=off, 1=on	`1`
`-downsample`	0=off, 1=on	`0`
`-custom`	0=off, 1=on	`1`

Commonly Used Optional Parameters

Parameter	Default	Description
`-algorithm`	`dfs`	Pathfinding algorithm: `dfs` or `sweep_line`
`-output_dir`	`outputs`	Output directory
`-flow`	`0.127`	Volumetric flow rate (mm^3/s)
`-print_speed`	`1.0`	Print speed (mm/s)
`-tolerance`	`0`	Tolerance amount (mm)

`-scale_factor`	`1.0`	Network scale multiplier
`-num_overlap`	`0`	Overlap nodes for gap closure

See the README for the complete list of 35+ optional parameters.

4.3 Mode 3: Docker

1. Build and start the container (if not already running):

```
docker build -t xcavate .
docker run -p 8501:8501 -v $(pwd)/outputs:/app/outputs xcavate
```

2. Open **http://localhost:8501** in your browser.
3. Follow the same steps as **Mode 1 (Web GUI)** above. All functionality is identical.
4. To use pre-written custom G-code template files, mount the `inputs/custom/` folder:

```
docker run -p 8501:8501 \
  -v $(pwd)/inputs/custom:/app/inputs/custom \
  -v $(pwd)/outputs:/app/outputs \
  xcavate
```

5. Output files are accessible in the mounted `outputs/` directory or via the download buttons.

5 Part E: Inspect Output Files

After execution completes, X-CAVATE produces the following outputs.

5.1 E.1 G-Code Files

Located in `outputs/gcode/`:

File	Description
`gcode_SM_pressure.txt`	Single-material G-code
`gcode_MM_pressure.txt`	Multimaterial G-code (if enabled)

The filename suffix matches your selected printer type (`pressure`, `positive_ink`, or `aerotech`).

5.2 E.2 Coordinate Files

Located in `outputs/graph/`:

File	Description
`SM_x_coords.txt`	Single-material x-coordinates per print pass
`SM_y_coords.txt`	Single-material y-coordinates per print pass

`SM_z_coords.txt`	Single-material z-coordinates per print pass
`SM_combined.txt`	Combined x, y, z per print pass
`MM_*.txt`	Multimaterial equivalents (if enabled)
`graph.txt`	Full adjacency graph
`special_nodes.txt`	Branchpoints, endpoints, inlets, outlets

5.3 E.3 Visualizations

Located in `outputs/plots/`:

File	Description
`network_SM.html`	Interactive 3D plot of single-material passes
`network_MM.html`	Interactive 3D plot of multimaterial passes

Open the `.html` files in any web browser to rotate, zoom, and inspect the print paths.

5.4 E.4 Changelog

The file `outputs/changelog.txt` contains a log of all gap closure operations performed during execution.

6 Troubleshooting

Problem	Solution
`Runtime instance already exists` (GUI)	A stale Streamlit process is running. Kill it: `pkill -f streamlit`, then restart.
Inlet/outlet matching warnings	The inlet/outlet coordinates are approximate. X-CAVATE matches to the nearest network point and logs the distance. Verify the matches are reasonable.
Pipeline hangs on large networks	Reduce the number of coordinates per vessel to 100 or fewer, or enable downsampling.
`ModuleNotFoundError`	Run `pip install ".[gui]"` to install all dependencies.
Docker container won't start	Ensure Docker Desktop is running and port 8501 is not in use.

7 Quick Reference: Mode Comparison

Feature	GUI	CLI	Docker
Python required	Yes	Yes	No
Custom G-code via files	No (use GUI editor)	Yes (`inputs/custom/`)	Optional (mount or GUI)
Custom G-code via editor	Yes	No	Yes

Interactive 3D plots	In-browser	Open `.html` files	In-browser
File download	Download buttons	Files in `outputs/`	Buttons + mounted volume
Best for	New users	Scripting, batch jobs	Reproducibility, sharing